CASE STUDY: Improving Digital KYC Conversion for HDFC Bank

Innovation Sprint 2025 – Digital KYC: Reduce Drop-Off, Lift Conversion

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1. Introduction

As banking rapidly shifts toward digital-first onboarding, seamless KYC completion becomes essential for customer acquisition. However, digital KYC journeys continue to suffer from high abandonment rates due to user confusion, network constraints, poor image quality, and delayed verification.

According to the HDFC Bank problem brief, the most significant drop-offs occur during:

- Document upload
- Document scanning
- KYC verification
- Real-time photo capture

This case study uses:
(a) the official scenario brief & guidelines provided by HDFC, and
(b) a 400-row session dataset (shared in this project),
to analyse patterns, uncover root causes, and propose a redesigned KYC journey that improves completion rate, reduces effort, and maintains regulatory compliance.

2. Objectives

- Reduce abandonment across the digital KYC funnel
- Improve per-step TAT to <15–20 seconds
- Reduce resubmissions & manual verifications
- Increase scan & upload success rate
- Maintain full compliance with RBI, V-CIP, CKYC regulations
- Improve youth/new-to-bank onboarding experience

3. Problem Context

Current KYC Journey

1. Select document type
2. Scan the chosen ID document
3. Upload the scanned image
4. Upload a passport-size photo
5. Capture real-time selfie
6. OCR → KYC verification → Approval

Observed Issues

Despite a structured journey, the platform suffers due to:

- Duplicate KYC entries
- Slow server response (>20s)
- Poor-quality document scanning
- Confusing UI (especially for first-time users)
- Users hitting the 3-attempt maximum limit

4. Regulatory Framework

RBI KYC Master Directions (2016–2023)

- Aadhaar + PAN mandatory for digital onboarding
- Mandatory liveness detection
- Geo-tagging for V-CIP
- Clear OVD capture

HDFC Document Guidelines

- Accepted: Aadhaar, PAN, Passport, Voter ID
- Photo must match real-time selfie
- Passport must have ≥ 6 months validity
- Each stage allows 3 attempts

These guidelines dictate the boundaries of the redesigned solution.

5. **Funnel Performance & Drop-Off Analysis**

Stage	Failure Rate	Key Causes
Select Document	15%	Confusing UI
Scan Document	35%	Blurry scans, glare, angle issues
Upload Document	25%	Network issues, OCR failures
KYC Check	15%	Duplicate/incorrect PAN/Aadhaar
KYC Approval	10%	Selfie mismatch

Overall Completion: ~32%
Biggest drop-off: Document Scan (35%)

6. Session-Level Dataset Insights (400 Rows Provided)

This strengthens the case study with real user behaviour.

6.1 Most Frequent Error Messages

From the 400 rows, the following errors dominate:

- "Please scan the correct document" (very frequent)
- "Please upload the correct selected document"
- "KYC data not found; please upload correct KYC"
- "KYC document already exists"
- "Please select correct document type"
- "Maximum tries exceeded"
- "Proceed to Document Upload"

6.2 Attempt Count Patterns

- Majority of failures occur at Attempt Count = 2 or 3
- Several rows show Attempt Count = 4, triggering hard lockouts:
    - "Maximum tries exceeded"
    - "Maximum upload tries exceeded"

6.3 Time-Taken Patterns

- Many rows show time 25–45 seconds, which exceeds the target (<20s).
- Indicates server delay, image-processing latency, or upload slowness.

6.4 High-Failure Stages Based on Data

From frequency analysis:

1. Document Scan → Most failing stage
2. Upload Document → Second highest failures
3. KYC Check → High 60% failure in many rows
4. Select Document Type → Consistent 10–15% failure
5. KYC Approved → 0% failure, minimal entries

7. ROOT CAUSE ANALYSIS

A. UX Barriers

- Users unsure how to position the document
- No real-time guidance (brightness, angle, glare)
- Dropdown for doc selection is confusing
- Elderly/rural users take longer → hit timeouts
- No multilingual instructions for scanning

B. Technical Issues

- Sequential processing increases latency
- OCR failures due to poor image quality
- Large files cause slow uploads
- No autosave → user restarts entire flow after errors

C. Device Constraints

- Low-end Android devices struggle with image processing
- Weak 2G/low-bandwidth areas increase upload time
- No offline-first capture mode

D. Process & Policy

- Hard lockout after 3 attempts
- Duplicate KYC checks occur too late
- Inadequate error clarity

8. Proposed UX Redesign

8.1 Simplified Document Selection

- Card-based layout instead of dropdown
- Sample images + clear microcopy
- Highlight mandatory documents (PAN + Aadhaar)

8.2 Smart Camera Capture Flow

- Auto-edge detection
- Auto-capture when steady
- On-screen hints (“Reduce glare”, “Hold steady”)
- On-device compression → 70% smaller files

8.3 Multilingual Support

Languages: EN, HI, AS, BN, MR
Addresses regional accessibility requirements.

8.4 Improved Error Messaging

Before:

“Invalid document. Try again.”

After:

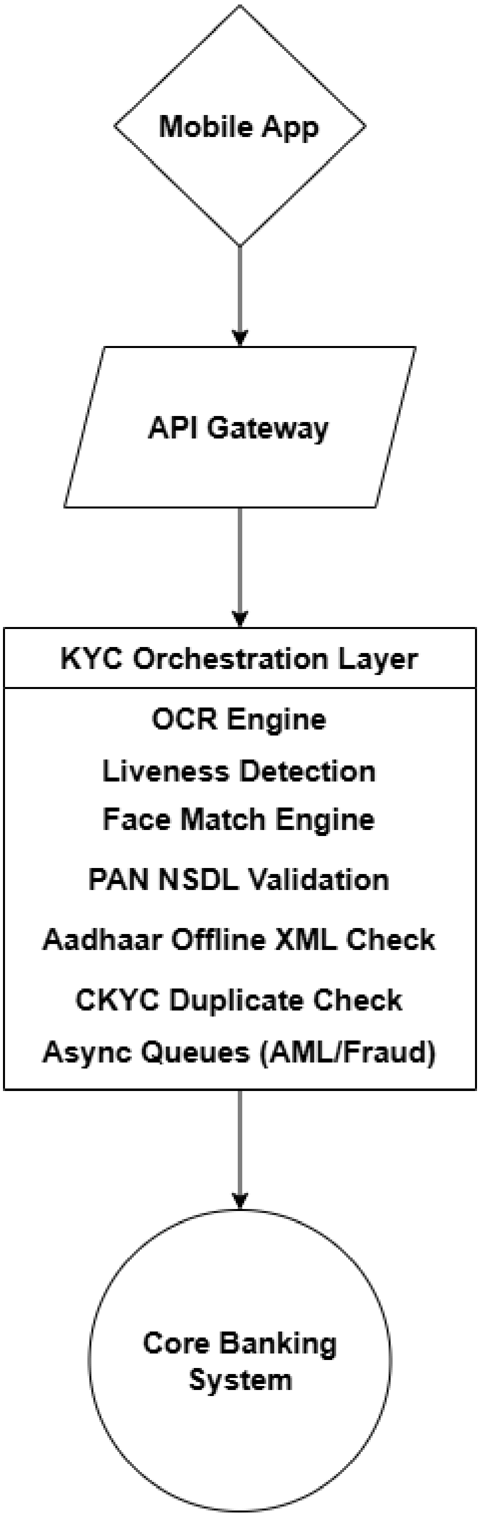
“We couldn’t read your PAN card due to glare. Move to brighter light and align within the frame.”

8.5 Reduced Steps

- Auto-fill using OCR
- Merge redundant fields
- Remove repeated confirmation screens

9. Technical Architecture Improvements

Modern Orchestration Layer:



Key Enhancements:

- Parallel processing for speed
- Resumable uploads
- Autosave at each step
- Device fingerprinting
- On-device compression
- Vendor-agnostic OCR

10. Security, Fraud & Privacy Controls

Security

- TLS 1.3
- AES-256 storage encryption
- Masking of PAN/Aadhaar
- Time-bound retention

Fraud Prevention

- Liveness detection
- Anti-spoofing
- Face-match threshold
- Device fingerprinting

Privacy

- Explicit consent
- PII minimization
- Pseudonymized logs

11. KPI Targets & Expected Impact

Metric	Current	Target
Scan Success	65%	85%
Upload Success	75%	90%
Verification Success	85%	90%
Overall Completion Rate	32%	55–60%

Business Impact

- 2× more successful onboardings
- 40% reduction in customer acquisition cost
- 30% fewer manual re-verifications
- Higher satisfaction (clear UI, faster TAT)

12. Future Roadmap

Short Term (0–3 months)

- Duplicate document warnings
- Interactive scanning guidance overlays

Medium Term (3–6 months)

- On-device OCR
- AI-based document quality scoring

Long Term (6–12 months)

- Fully automated V-CIP
- Voice-led onboarding
- Federated identity support (future Aadhaar APIs, DigiLocker)

13. Conclusion

The proposed redesign provides a scalable, compliant, and user-centric digital KYC framework. By combining UX improvements, back-end performance enhancements, multilingual support, and intelligent scanning, the new journey significantly reduces drop-offs, enhances customer trust, and improves operational efficiency.